Reg. No.: 22BAI1266

Name : AFZAL

VIT

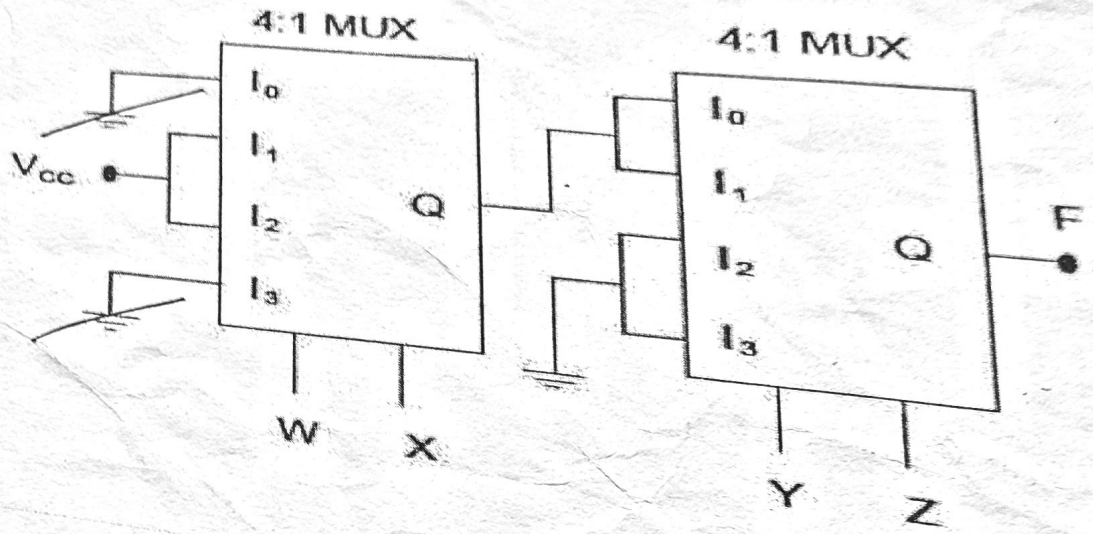
Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

Continuous Assessment Test - 1 (CAT 1) – March 2023

Programme	:	B. Tech CSE & Splns	Semester	:	Winter 22-23
Course	:	Digital Systems Design	Code	:	BECE102L
			Slot	:	B1+TB1
Faculty	:	Dr. Ashok Mondal Dr. Deepa T Prof. Deepa M Dr. Meera P S Dr. Sriramalakshmi P Dr. Subbulekshmi D	Class Nbr	:	CH2022232300193 CH2022232300530 CH2022232300191 CH2022232300559 CH2022232300535 CH2022232300526
Time	:	1.5 hours	Max. Marks	:	50

Answer ALL Questions

Q.No.	Sub. Sec.	Question Description	Marks
1		Simplify the following expressions using Boolean algebra. a) $Y = (M+N)(\overline{M}+P)(\overline{N}+P)$ b) $\overline{\overline{AB}(CD+\overline{E}F)(\overline{AB}+\overline{CD})}$ c) $(B+BC)(B+\overline{B}C)(B+D)$ d) $Y = A\overline{B}C + (\overline{B}+\overline{C})(\overline{B}+\overline{D}) + \overline{A+C+D}$	10
2		Reduce the logic function F(A,B,C,D) = ∑m (1,2,3,4,7,11,13) + d (9,15) using K- map in SOP form. Implement the minimized expression using only NAND gates.	10
3		Design a digital system whose output is defined as logically low if the 4 bit input binary number is a multiple of 3; otherwise, the output will be logically high. The output is defined if and only if the input binary number is greater than 2. Draw the logical diagram for the simplified expression for the above system.	10
4		For the Boolean expression, F(A,B,C,D) = (A+B'+D) (A'+B+C') (B+C+D'), a) draw the truth table and mark the outputs correctly. Realize the expression using b) 3:8 decoders with active high enable. c) 8:1 multiplexer and external gates (Select lines: A, B, C). d) 4: 1 multiplexer and external gates (Select lines: A, B).	15

5 For the circuit given below, find the expression for F. 5